

2026 首届 OPENVELA AI 硬件开发者大赛 · PROJECT INTRODUCTION



VelaPaw

An on-device AI multi-pet smart feeder that recognizes which pet is at the bowl and feeds it its own scheduled meal — no cloud, no phone, no account.

Team CircuitForge (contest2026_043)

Track AI Hardware Product Innovation / AI 硬件产品创新

Platform openVela (NuttX) on Waveshare ESP32-S3-Touch-LCD-3.5-C

Repository github.com/open-vela/contest2026_043_CircuitForge

01 The Problem, and the Idea

In a multi-pet home, a normal timed feeder can't tell the animals apart — so the greedy one eats everyone's portion, and per-pet diet or health tracking is impossible. VelaPaw solves this the way a person would: it looks at the animal, identifies it, and feeds it only its own ration, at its own mealtimes.

The core mechanism. A pet is fed only when it is *recognized at the bowl* **and** one of *its* meals is due and not yet eaten. Recognition alone would let a greedy pet eat all day; a timer alone would feed whichever animal happened to be standing there. Requiring both closes the gap that every timer-only feeder on the market leaves open.

Fully on-device

Camera, neural networks, clock, and dispenser are all self-contained on one ESP32-S3. It keeps working with the internet down and never sends images of your home anywhere.

No retraining to add a pet

Recognition is open-set metric learning, not a fixed classifier — enrolling a new pet is a runtime operation (a few photos), not a model update.

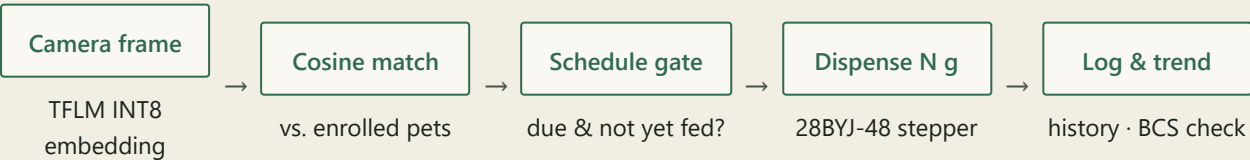
Health signal, not just a bowl

A second on-device model estimates body condition (under/ideal/over), and feeding-history analytics raise a sudden-appetite-drop alert — an early illness sign.

Hands-free scheduling

A spoken MEAL→HOUR→CONFIRM voice dialog, driven by an on-device keyword-spotting model, sets a pet's feeding schedule without touching the screen.

02 How It Works — Camera to Kibble



COMPONENT	DETAIL
Identity model	MobileNetV3-Small backbone + spatial-attention (TSFM) module, INT8, 128-D L2-normalized embedding, 128×128 input

COMPONENT	DETAIL
Body-condition model	Compact 3-class (under / ideal / over) INT8 CNN
Runtime	TensorFlow Lite for Microcontrollers (apps/mlearning/tflite-micro), two independent MicroInterpreters with PSRAM tensor arenas
Acceleration	ESP-NN SIMD kernels on the Xtensa LX7 — ≈ 1.3 s / inference (vs. ≈ 6.75 s on reference kernels)
Model storage	Flash-loaded at boot: 1.44 MB identity model @ 0x600000, 626 KB BCS model @ 0x760000 — too large to memory-map, read into PSRAM via a dedicated boot-time thread

Both models are trained off-device in host/ (TensorFlow/Keras → INT8 post-training quantization → .tflite); the identity model additionally uses an ArcFace metric-learning head so embeddings separate cleanly by identity.

03 Engineering Depth & Technical Difficulty

VelaPaw is not a thin app on top of a stock board support package — it exercises openVela across the stack, with custom board bring-up written from scratch for every peripheral the product needs.

- **Display** — ST7796 touch LCD over hardware SPI2 + GDMA (40 MHz)
- **Camera** — OV5640 via the ESP32-S3 LCD_CAM peripheral with a GDMA receive channel, configured over SCCB/I²C, relocated onto a 30 cm FFC + coupler so the camera mounts independently of the board
- **Real-time clock** — PCF85063 over I²C via raw register access (deliberately not CONFIG_RTC, which would otherwise kill the USB-CDC console)
- **Feeder actuator** — 28BYJ-48 stepper through a ULN2003 driver on four GPIOs, coil sequence stepped in software
- **Audio** — the ES8311 codec's private I²S bus drives the onboard mic/speaker for voice-settable schedules
- **AI runtime** — apps/mllearning/tflite-micro linked directly into the app, running two concurrent interpreters

Case study: the hardware-SPI display bug

SYMPTOM	Full-screen SPI writes rendered; any partial-window write never appeared — identical bytes over bit-banged GPIO worked perfectly. Survived ~60 blind build cycles.
METHOD	Logic analyzer (sigrok/PulseView, FX2 clone) mirrored onto spare header GPIOs via the GPIO matrix, validated first against known square waves before trusting any real capture.
FINDING	The ESP32 was emitting a byte-perfect CASET/RASET/RAMWR stream — the peripheral was innocent.
ROOT CAUSE	The panel's chip-select is hardwired low, so it can never re-sync its bit counter on a CS edge. Bit-banged init handed the pads to the SPI peripheral mid-stream, glitching the clock by one edge and desyncing the panel permanently.
FIX	Hand the pads to the SPI peripheral first, hardware-reset the panel, then run all init over SPI — re-syncing the panel under SPI ownership from the start.

Performance engineering

5×

INFERENCE SPEED-UP
ESP-NN SIMD KERNELS · 6.75S → 1.3S

4×

UI REPAINT SPEED-UP
SPI+DMA · 246MS → 61MS FULL REPAINT

04 Hardware & Industrial Design

Board choice — same SoC, right peripherals

The Waveshare **ESP32-S3-Touch-LCD-3.5-C** carries the same ESP32-S3 as the contest-provided ESP32-S3-EYE, so the openVela port, on-device inference, and every driver are identical work on that chip. What VelaPaw additionally needs — a capacitive touch display for enrollment/scheduling/trends, and a real-time clock so schedules survive power loss — comes integrated on this board rather than bolted on: 320×480 ST7796 touch LCD, OV5640 camera, and PCF85063 RTC in a single carrier.

Feeder mechanism

A 28BYJ-48 stepper-driven paddle-rotor dispenser (via a ULN2003 driver): a vaned rotor meters a fixed pocket of kibble per step, so portion size depends on rotation count, not on how full the hopper is. Fully parametric OpenSCAD model, printable STLs, and BOM are included in hardware/.

Camera placement

The OV5640 is unplugged from the board and run out on a 30 cm ribbon + 1:1 coupler to its own cam_mast yoke positioned in front of the bowl — so the screen faces the owner while the lens watches the pet.

05 AI-Assisted Development

VelaPaw was built end-to-end with AI-assisted (AI Coding) development, across the full lifecycle — not just code generation.

Requirements & design

Shaping the recognition-gated feeding model, the open-set enrollment approach, and the health-monitoring analytics.

Firmware & drivers

Bringing up display, camera, RTC and stepper feeder on the ESP32-S3; wiring TensorFlow Lite Micro + ESP-NN onto the Xtensa LX7.

Hard debugging

The SPI display case study above — instrumented, measured, and root-caused with AI assistance driving the logic-analyzer investigation.

Product iteration

The full pet lifecycle — enroll, recognize, feed, edit, delete, photos — built and tested on real hardware.

Reusable Skills produced

Development used a set of purpose-built Claude Code Skills, committed for reuse rather than one-off prompts: board bring-up / build automation, driver code review, PR submission, and AI-coding session-log collection.

Session logs

Every AI-coding session is exported to `logs/KingMbewe/` as required, organized by date with a per-member manifest — the full development history, not a curated excerpt.

06 Market & Value

Multi-pet households are a large and specific segment underserved by today's smart feeders, which are almost universally single-bowl, timer-based, and cloud/app-dependent.

The category is already validated. In March 2026, Xiaomi shipped the **Mijia Smart Pet Feeder 2 (Visual Edition)** — a mainstream, mass-market device (¥449–549, ≈\$65–80) that added a 5MP camera specifically to bring AI vision into feeding: 150° field of view, 940nm night vision, and on-bowl activity

detection. That's independent, at-scale confirmation of VelaPaw's founding bet — that a camera belongs at the feeder — from the company best positioned to know what pet owners will actually buy.

VelaPaw is best read as an extension of that trajectory, not a green-field pitch: the same instinct — put vision at the bowl — carried one layer further, from watching a bowl to knowing whose bowl it is.

CAPABILITY	XIAOMI MIJIA FEEDER 2 (VISUAL EDITION)	VELAPAW
What the camera does	Detects a pet at the bowl and logs its eating activity	Identifies <i>which enrolled pet</i> is at the bowl, by face
Multi-pet portioning	One schedule, dispensed to whichever animal is present	Feeding gated on both the pet's identity and its own schedule
Data path	Mi Home app; local microSD or paid cloud storage	Fully on-device — no app, no account, nothing leaves the box
Health signal	Eating-activity log	Dedicated body-condition scoring model + appetite-drop alerts
Offline operation	App-connected features depend on network	Full function with no internet at all

Xiaomi specs per its official Mijia Smart Pet Feeder 2 (Visual Edition) launch, March 2026 — Youpin crowdfunding ¥449 / retail ¥549, China-exclusive.

Differentiation

Closes the one gap a bowl-facing camera like Xiaomi's doesn't: telling the pets apart. Identification is by face, not an RFID collar tag — no extra hardware to attach to the animal.

Privacy & reliability

No cloud dependency, no account, no subscription, and no images ever leave the device — a genuine differentiator against app-gated competitors, and it keeps working through an internet or Wi-Fi outage.

Path to a product

The feeder mechanism and enclosure are already fully parametric (OpenSCAD + STL + BOM), a direct head start toward a production-ready custom PCB and injection-molded housing.

Platform value

Beyond the product itself, VelaPaw is a concrete, shipped reference for what an openVela-native AI Agent hardware product looks like — on-device recognition, voice, and scheduling, with no cloud in the loop.

From Prototype to Product

The current build is deliberately an **engineering prototype**, not a cost-optimized one: a general-purpose Waveshare dev board, a generic ULN2003 stepper-driver module, and jumper-wired

connections. That's an intentional trade — it kept the contest timeline focused on the parts that carried real technical risk (on-device recognition, the SPI display fix, driver bring-up), all of which are now done and verified on real hardware.

The path from here to a sellable unit is standard hardware productization, not open R&D:

- **Custom PCB** — fold the ESP32-S3, display driver, RTC, and stepper driver onto one board, replacing the dev board + breakout module + loose wiring. Layout work against an already-working reference design, not a new circuit to invent.
- **Injection-molded enclosure** — the structural parts are already parametric OpenSCAD, printed today through a print-on-demand service ([hardware/stl_j1cpcb/](#)); the same geometry is the direct input to injection-mold tooling once volume justifies it.
- **Certification** — a camera-, microphone-, and Wi-Fi-equipped consumer device needs standard regulatory clearance (e.g. China 3C/SRRC, or CE/FCC elsewhere) before retail sale — a bounded, well-understood step, not a technical unknown.
- **Volume sourcing** — the stepper, driver, and RTC are all commodity components already priced at hobbyist quantities; cost drops predictably as volume increases, unlike the AI models and firmware, which are the actual hard part and are already built.

In other words: the parts of this project that carry real technical risk — the recognition pipeline, the display driver, the feeder mechanism — are done. What's left to reach a product is execution, not invention.

07 Current Status & Roadmap

Working on hardware today

- ✓ On-device recognition and body-condition scoring
- ✓ Recognition-gated 3-meals/day scheduling with per-slot skip-meal toggle
- ✓ Portion & daily-limit control
- ✓ Feeding history, trends, and appetite-drop alerts
- ✓ Hardware RTC keeping schedules accurate across power loss
- ✓ Full pet lifecycle with persistent photos; auto-start straight into the UI
- ✓ Hardware-SPI + DMA display; bilingual EN/中文 UI with a live toggle
- ✓ Spoken MEAL→HOUR→CONFIRM scheduling dialog, proven end to end on hardware
- ✓ 28BYJ-48 stepper feeder turning on real GPIO wiring
- ✓ OV5640 relocated onto a 30 cm FFC, confirmed working through the mast mount
- ✓ 3D-printed enclosure fully assembled — stepper dispenser and relocated camera mounted into the printed structure

In progress

- An intermittent I²S capture stall on the voice path, not yet fully root-caused